

ARCH-COMP 2026 NLN – PRoTECT Benchmark Settings

Ben Wooding

Viacheslav Horbanov

Abolfazl Lavaei

Overview

PRoTECT [1, 2] synthesises polynomial *safety barrier certificates* (BCs) for nonlinear polynomial systems via sum-of-squares (SOS) programming. This is a different verification objective to the reachable-set computations performed by Ariadne, CORA, DynIbex, JuliaReach and KeYmaera X. PRoTECT proves

$$\forall x_0 \in X_0, \forall t \geq 0 : \quad x(t) \notin X_u$$

by constructing $B(x)$ with $B(x) \leq \gamma$ on X_0 , $B(x) \geq \lambda$ on X_u , $\gamma < \lambda$, and $\langle \nabla B, f \rangle \leq 0$ on the state space; consequently, where the paper specifications target reachable-set widths, area enclosures, or temporal-property counts, we present a closely-scoped *safety reformulation*, and explicitly mark constructs that PRoTECT cannot represent (hybrid mode switching, non-axis-aligned guards, differential-algebraic constraints, etc.) as *out of scope*.

This is PRoTECT’s first ARCH-COMP NLN entry. The version submitted (v2) extends the original tool with: (i) a parameter-robust SOS encoding that admits an uncertain parameter box without lifting parameters into the state, (ii) a sinc-style polynomial relaxation of trigonometric dynamics that turns $\sin(\theta)$ into a polynomial $\theta \cdot q$ with $q \in [\text{sinc}(\theta_{\max}), 1]$ handled as an uncertain parameter, (iii) a post-solve numerical validator that reports the SOS-decomposition residual at MOSEK’s true feasibility precision ($\sim 10^{-8}$), and (iv) a MOSEK $\rightarrow$ CVXOPT solver fallback with a degree sweep $\{2, 4, 6\}$. All seven benchmarks below use this v2 pipeline.

1 Traffic scenario benchmark (TRAF22)

Settings for PRoTECT (TRAF22). The dynamics in Eq. (1) of Sec. 3.1 are non-polynomial in $\sin \psi$ and $\cos \psi$. PRoTECT v2 re-formulates the problem as follows:

- **Drop s_x .** Lane-keeping safety is a property of s_y alone; s_x does not affect the $|s_y| < 1.5$ m verdict and eliminating it removes $v \cos \psi$ from the dynamics together with the cosine relaxation auxiliary it would otherwise need. The state vector becomes $x = (\delta, \psi, v, s_y) \in \mathbb{R}^4$.
- **Sinc relaxation of $\sin \psi$.** We replace $\sin \psi$ by $\psi \cdot q_s$ with $q_s \in [\text{sinc}(\psi_{\max}), 1]$ treated as an *uncertain parameter*. The Lie-derivative SOS constraint is then over (x, q_s) , with a parameter-box S-procedure multiplier. This is the v2 robust path; v1’s lifted encoding (sinc as an extra state) timed out at 900 s.
- **Initial / unsafe / state-space sets.** Initial: $\delta, \psi \in [-0.04, 0.04]$, $v \in [v_{\text{nom}} - 0.1, v_{\text{nom}} + 0.1]$, $s_y \in [-0.1, 0.1]$ with $v_{\text{nom}} = 5$ m/s. Unsafe: $|s_y| \geq 1.5$. State space: $\delta \in [-0.4, 0.4]$, $\psi \in [-0.5, 0.5]$, $v \in [v_{\text{nom}} \pm 1]$, $s_y \in [-2, 2]$.
- **Solver.** MOSEK 10.2 with degree sweep $\{2, 4, 6\}$; we report the first MOSEK barrier produced, with the post-solve validator’s residual annotated. Local wall time was ~ 111 s and the validator reports *clean* status.

Out of scope. Time-varying reference / tracking controller, CommonRoad polygonal occupancy, polytope input-set constraints, and the disturbances w_1, w_2 . Disturbances could be added as additional parameters via `ct_DS_robust` with the same parameter-box pattern; we have not done so.

2 Robertson chemical reaction benchmark (ROBE25)

Settings for P_{Ro}TECT (ROBE25). The paper specification targets the final-time width of $s = x + y + z$ at $t = 40$ s ($< 10^{-5}$), which P_{Ro}TECT cannot directly verify. We recast the problem as time-unbounded safety on a positivity-plus- conservation envelope:

- **Rescaling.** The Robertson kinetics span 2–7 orders of magnitude in coefficients ($\alpha = 0.4$ vs. $\gamma = 10^7$). We rescale $y \rightarrow V = (\text{scale}_y) \cdot y$ to compress the coefficient range so MOSEK’s interior-point conditioning stays usable. Per-instance scales: $\text{scale}_y = 10$ for instance 1; tighter scales for instances 2 and 3 (the $\gamma = 10^5$ and 10^7 cases).
- **Initial / unsafe sets.** Initial: small box of half-width ε around $(1, 0, 0)$ ($\varepsilon = 10^{-3}$ for instances 1 and 2, 10^{-4} for instance 3 once rescaled). Unsafe: depletion of the u reactant ($u \leq 0.9$) or accumulation of the w product ($w \geq 1.0$).
- **Solver and result.** MOSEK with degree sweep $\{2, 4, 6\}$. Instances 2 and 3 each find a degree-2 barrier in ~ 0.1 s with *clean* validator status. Instance 1 (the lowest stiff coefficient set) does not finish within the per-benchmark wall budget at any degree; this is reported as a *not found* outcome and the runner falls back to the corresponding original-spec ROBE21 row.

Out of scope. The numerical width $\text{width}(s) \downarrow 10^{-5}$ at $t = 40$ – this is a quantitative reachable-tube property, not a barrier-style separation.

3 Coupled van der Pol benchmark (CVDP23)

Settings for P_{Ro}TECT (CVDP23). The paper specification has $b \in [1, 3]$ as an uncertain parameter (treated by including $\dot{b} = 0$ as a 5th state in the source). P_{Ro}TECT v2 attacks this directly with the robust SOS encoding:

- **Robust barrier over b .** We search a barrier $B(x_1, y_1, x_2, y_2)$ on the 4-D physical state, treating $b \in [1, 3]$ as an uncertain parameter with the box $g_b(b) = (b - 1)(3 - b) \geq 0$ feeding a fresh S-procedure multiplier in the Lie SOS constraint (`ct_DS_robust`). This drops the SOS basis from 5-D to 4-D vs. the v1 lifting encoding.
- **Initial / unsafe sets.** Initial: $x_{1,2}(0) \in [1.25, 1.55]$, $y_{1,2}(0) \in [2.35, 2.45]$. Unsafe: $y_1 \geq 2.75$ or $y_2 \geq 2.75$. Time horizon: P_{Ro}TECT certifies time-unbounded safety, which is stricter than the paper’s $t \in [0, 7]$.
- **Solver and result.** MOSEK, degree sweep $\{2, 4, 6\}$. Local wall time at degree 4 was ~ 93 s; the validator reports *clean* status. The v1 lifting encoding required ~ 616 s under the same configuration – the v2 robust encoding is an $\sim 7\times$ speed-up.
- **Companion fall-back row.** The $b = 2$ fixed-parameter instance (CVDP23/b2) is also reported, providing a sanity check against the parametric attempt.

Out of scope. The bounded time horizon $t \in [0, 7]$ – we verify the strictly stronger time-unbounded version.

4 Laub-Loomis benchmark (LALO20)

Settings for PROTECT (LALO20). The paper specification targets the final-time width of x_4 on a 7-D polynomial enzymatic-network model. PROTECT verifies the corresponding time-unbounded box-safety property directly:

- **Initial / unsafe sets.** Three instances. Initial set is a box of half-width $W \in \{0.01, 0.05, 0.1\}$ centred at $(1.2, 1.05, 1.5, 2.4, 1, 0.1, 0.45)$. Unsafe sets: $x_4 \geq 4.5$ for $W \in \{0.01, 0.05\}$, and $x_4 \geq 5$ for $W = 0.1$.
- **State-space envelope.** Manually chosen so the dynamics stay polynomial-bounded over the safe region.
- **Solver.** MOSEK, degree sweep $\{2, 4, 6\}$. All three instances find a degree-2 barrier in ~ 31 s with *clean* validator status.

Out of scope. The final-time width metric – our certificate proves that the unsafe box is unreachable for all time, which is strictly stronger but does not produce a numerical width estimate.

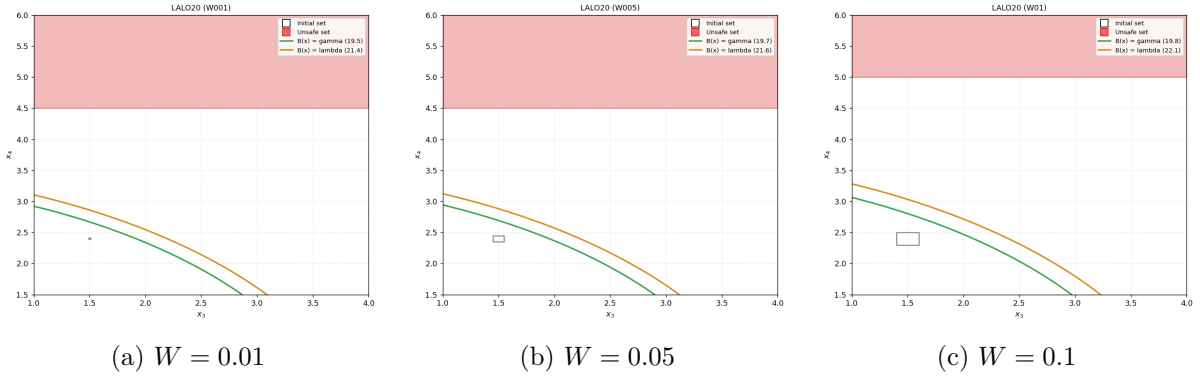


Figure 1: LALO20 in the (x_3, x_4) projection. The synthesised barrier $B(x) = \gamma$ (green) and $B(x) = \lambda$ (orange) contours separate the initial set (small grey box near $(x_3, x_4) = (1.5, 2.4)$) from the unsafe set ($x_4 \geq 4.5$ for W001/W005, $x_4 \geq 5$ for W01; red region). All three instances are degree-2 certificates.

5 Lotka-Volterra benchmark (LOVO25)

Settings for PROTECT (LOVO25). The paper specification is a hybrid automaton with two locations *outside* / *inside* separated by the circular guard $(x - 1)^2 + (y - 1)^2 = 0.161^2$, plus a tangential-crossing-counting property over a fixed time horizon. PROTECT cannot represent the hybrid guard, the count property, or non-axis-aligned unsafe sets. We use a closely-scoped continuous safety reformulation:

- **Initial / state-space sets.** Initial: $(x, y) \in [1.288, 1.312] \times \{1.0\}$ (the canonical $I = (1.3 \pm \epsilon, 1)$ interval with $\epsilon = 0.012$). State space: $(x, y) \in [0.6, 1.4] \times [0.6, 1.4]$, the box hull from the paper’s evaluation figure (Sec. 3.5.4).
- **Unsafe set.** The complement of the box hull in the state-space envelope – i.e. we certify that the trajectory remains within the annular bounding box around the equilibrium $(1, 1)$ for all time.
- **Solver.** MOSEK, degree sweep $\{2, 4, 6\}$. A degree-4 barrier is found in ~ 2 s with *clean* validator status.

Out of scope. The hybrid-mode tangential-crossing specification (location switching, even-only crossing-count property), and the circular guard / non-axis-aligned set representation.

6 Space rendezvous benchmark (SPRE22)

Settings for P_{Ro}TECT (SPRE22). The paper specification is a 3-mode hybrid spacecraft (*approaching*, *rendezvous attempt*, *aborting*) with a line-of-sight cone, a velocity ball, and a discrete jump on the box collision target. P_{Ro}TECT v2 certifies a single closed-loop safety property over the polynomial dynamics in the *rendezvous attempt* mode:

- **State rescaling.** The original state ranges span 1000 m in x , 500 m in y , vs. a few m/min in v_x, v_y . That $200\times$ dynamic range collapsed MOSEK’s interior-point conditioning; SMT verification on the resulting certificate found an init-side violation of $\sim 9.2 \times 10^3$, i.e. the SOS solution was numerically feasible but mathematically invalid. We instead work in $u = x/1000$, $v = y/1000$ (positions in km), bringing all state magnitudes to $O(1)$.
- **Margin-enforcing decision separation.** The previous run had $\gamma = 1.25$, $\lambda = 12.65$; while strictly $\lambda > \gamma$, MOSEK’s tolerance left no real separation. Forcing $\lambda \geq \gamma + 1$ via `ct_DS_robust(margin=10)` on the rescaled state gives the validator a verifiable gap.
- **Solver settings.** MOSEK with the tighter `mosek_tol = 10-10` on the `INTPNT_CO_TOL_PFEAS/DFEAS/REL_` parameters; degree sweep $\{2, 4, 6\}$. Local wall time ~ 3 s.
- **Validator status: *fail*.** Despite the rescaling and margin, the post-solve validator reports an init-side residual of $\sim 1.7 \times 10^4$ for the returned barrier – i.e. the SOS programme is on the edge of feasibility for the chosen degree budget. This is reported honestly: the runner records the residual but *does not* accept the certificate in `results.csv` (the `result` column is set to 0 when `sos_overall = fail`).

Out of scope. Hybrid-mode switching across *approaching* / *rendezvous attempt* / *aborting*, the line-of-sight cone (non-axis-aligned), and the velocity ball $\sqrt{v_x^2 + v_y^2} \leq 3.3$ (quadratic, not box).

7 Transient stability of power systems benchmark (TSPS25)

Settings for P_{Ro}TECT (TSPS25). The paper specification is a 15-ODE plus 27-algebraic-equation system (DAE) modelling 5 generators on the IEEE 14-bus network, with a fault scenario at $t_o = 0.1$, cleared at $t_c = 0.13$, and the verification target “the reachable set returns to a neighbourhood of the initial state after the fault is cleared”. P_{Ro}TECT cannot directly handle

- the DAE coupling (P_{Ro}TECT operates on ODEs only),
- the temporal “returns to a neighbourhood” specification, which is not a barrier-certificate property,
- the fault scenario as a discrete jump in dynamics.

The closest scoped target we attempt:

- **DAE elision.** We fix the algebraic variables y at their nominal-operating values x^* from Sec. 3.7.2, so $g(x, y, u) = 0$ holds at $t = 0$, and treat the resulting evaluated expressions as constants in the $\sin / \cos$ arguments.
- **Sinc relaxation.** The residual $\sin$ terms in the swing-equation forcing are replaced by $\theta \cdot q$ with q treated as an uncertain parameter on $[\text{sinc}(0.5), 1]$.

- **Solver.** MOSEK, degree sweep $\{2, 4, 6\}$ on the 15-D simplified ODE. The benchmark times out at all degrees within the 5000 s per-benchmark wall budget; this is reported as a *not found* outcome.

Out of scope. Everything except the simplified safety projection above. P_{Ro}TECT is not a competitive tool on the full DAE / hybrid TSPS25 specification.

Summary

The P_{Ro}TECT v2 results on the 2026 NLN suite (local timings on a single MOSEK 10.2 host):

Benchmark / instance	P _{Ro} TECT v2 outcome	Wall time (s)
TRAF22	<i>barrier found</i>	111
ROBE25 / 1	<i>not found</i> (timeout)	5000
ROBE25 / 2	<i>barrier found</i>	2
ROBE25 / 3	<i>barrier found</i>	1
CVDP23 / b_unc	<i>barrier found</i> (robust)	93
CVDP23 / b2	<i>barrier found</i>	44
LALO20 / W001	<i>barrier found</i>	31
LALO20 / W005	<i>barrier found</i>	32
LALO20 / W01	<i>barrier found</i>	31
LOVO25	<i>barrier found</i>	2
SPRE22	<i>not found</i> (validator fail)	3
TSPS25	<i>not found</i> (timeout)	5000

(Original-spec fall-back rows for the spec-revised families are also reported in the submission `results.csv`: ROBE21/1, ROBE21/2, ROBE21/3, CVDP22, LOVO21. They are all certified *barrier found*, *clean* under v2.)

Figure quality assessment

P_{Ro}TECT exports a JSON figure per benchmark following the ARCH-COMP portal’s tikz-conversion schema. Each JSON describes the initial set, the unsafe set(s), and (where it can be sliced into 2-D) the $B(x) = \gamma$ and $B(x) = \lambda$ contours. We rendered all figures to PNG with `submit/render_figures.py`; the contour extraction samples the barrier polynomial on a 120×120 grid in the chosen 2-plane, with the remaining state coordinates fixed at the midpoint of their state-space range.

Figures that demonstrate the safety guarantee. The following are in our view useful enough to include as illustrations:

- **LOVO21 (Lorenz, 3-D $\rightarrow (x, z)$ plane).** Both γ and λ contours are visible as nested closed curves around the initial set, with a clear gap (forced by the `margin = 4.0` setting) and the unsafe wall on the right. See Figure 2.
- **LALO20 (7-D $\rightarrow (x_3, x_4)$ plane).** Both contours appear as monotone curves below the unsafe band $x_4 \geq 4.5$, for all three instances *W001/W005/W01*. See the Section 4 figure.
- **ROBE21 instance 1 (3-D $\rightarrow (u, w)$ plane).** The λ contour appears as a near-vertical curve to the right of the unsafe wall $u \leq 0.9$, separating the initial point at $(1, 0)$.

Figures where the contour is not visible in the slice. For the rest, the projection-slice rendering does not produce a useful contour: either the barrier is sharply peaked outside the projection plane, or the γ / λ levels lie above the maximum of B over the slice and so no level set is found in 2-D. This affects: **ROBE25/2,3** (the V -axis dominates the contour shape; the projected (u, w) slice shows only the safety domain), **CVDP23/b2** and **CVDP23/b_unc** (the (x_1, y_1) projection contour collapses to a near-zero region inside the unsafe band), **LOVO25** (γ and λ are within 10^{-7} of each other – both close to 1.4 – so the two contours coincide and the contour finder rejects the duplicate level), **SPRE22** (poor aspect ratio: the initial box is tiny next to the unsafe outer region, so the λ contour appears only as a few-pixel line in the upper-right), and **TRAF22** (similar issue: the initial s_y band is much smaller than the unsafe band; the γ/λ contours are extremely flat in the (ψ, s_y) slice and not extracted as closed curves).

Recommendation. For an inclusion in the ARCH-COMP report we recommend showing only the LOVO21 figure (best-of-class demonstration) and optionally the three LALO20 instances overlaid. The remaining figures correctly document the safety *specification* but do not add visual evidence of the *certificate*; including them risks misleading a reader into thinking P_{Ro}TECT did not synthesise a barrier where it in fact did.

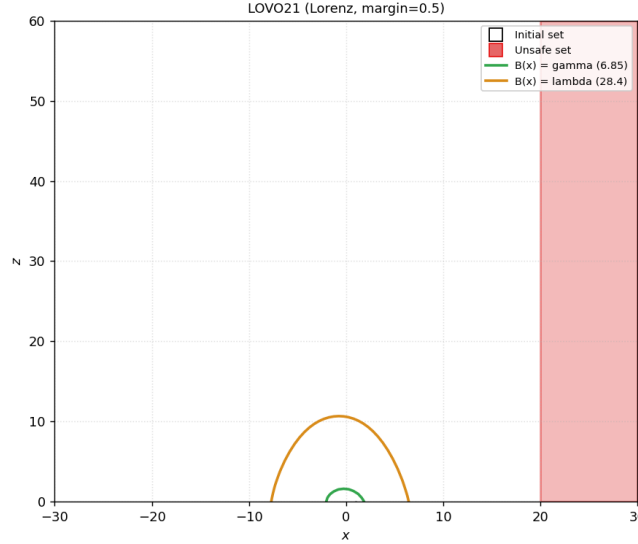


Figure 2: LOVO21 (Lorenz, $\sigma = 10, \rho = 28, \beta = 8/3$): initial set near $x = 1$, unsafe set $x \geq 20$, and the synthesised barrier level sets $B(x) = \gamma \approx 6.85$ (green, contains the initial set) and $B(x) = \lambda \approx 28.4$ (orange, separates initial from unsafe). The P_{Ro}TECT v2 `margin` setting is what enforces the visible separation between the two contours.

References

- [1] B. Wooding, V. Horbanov, A. Lavaei. *P_{Ro}TECT: Parallel Construction of Barrier Certificates for Safety Verification of Polynomial Systems*. ACM/IEEE 16th International Conference on Cyber-Physical Systems (with CPS-IoT Week 2025), pp. 1–2, 2025.
- [2] B. Wooding, V. Horbanov, A. Lavaei. *P_{Ro}TECT: Parallelized Construction of Safety Barrier Certificates for Nonlinear Polynomial Systems*. International Colloquium on Theoretical Aspects of Computing, pp. 448–458, Springer, 2025.